

MARTHA'S VINEYARD, MA, USA

WMO: 725066

Lat: 41.393N

Lon: 70.615W

Elev: 68

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 94724

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	14.3	-6.5	3.9	14.3	-1.9	5.0	17.7	31.2	31.6	27.9	32.0	10.2	330	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	84.2	72.8	81.6	71.7	79.4	70.6	75.7	80.9	74.4	79.0	73.0	77.1	9.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	126.9	78.6	72.8	121.8	77.1	71.9	118.2	75.9	39.2	81.4	37.9	79.1	36.6	77.3	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.0	23.4	20.3	DB	1.9	89.4	5.9	2.8	-2.4	91.5	-5.8	93.1	-9.1	94.7	-13.4	96.8	(4)
(5)			WB	0.9	78.1	5.6	1.6	-3.1	79.3	-6.4	80.2	-9.6	81.1	-13.6	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	31.6	31.9	37.4	46.1	55.7	63.7	70.3	69.5	63.2	53.6	45.1	37.3
	DBStd	15.44	9.79	8.07	7.75	6.30	6.38	6.02	5.41	5.62	6.67	7.71	8.24	8.64
	HDD50	2289	572	506	395	149	19	0	0	0	2	52	191	403
	HDD65	5745	1036	926	856	568	297	94	17	22	112	361	598	858
	CDD50	2488	1	1	5	32	194	411	628	604	397	162	43	10
	CDD65	469	0	0	0	1	7	55	180	162	57	7	0	0
	CDH74	2456	0	0	0	11	49	270	988	939	189	10	0	0
	CDH80	426	0	0	0	4	8	52	200	148	13	1	0	0
Wind	WSAvg	10.1	11.2	11.3	11.8	11.2	10.1	9.0	8.4	7.9	8.7	10.1	10.8	11.0
Precipitation	PrecAvg	44.8	3.4	3.2	4.4	3.8	3.5	3.3	2.7	3.5	4.0	3.9	4.1	4.6
	PrecMax	59.3	5.9	6.2	8.6	7.2	6.5	10.4	6.2	7.7	12.9	8.8	7.6	9.2
	PrecMin	39.0	1.7	1.6	0.8	1.5	1.0	0.4	0.5	0.1	0.9	0.7	0.7	1.5
	PrecStd	4.4	1.1	1.2	2.0	1.7	1.5	2.1	1.2	1.9	2.4	2.2	1.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	54.4	61.9	70.4	80.0	85.7	88.7	87.7	81.4	74.7	65.4	61.0
		MCWB	52.6	49.0	51.7	55.3	65.3	72.3	75.8	74.7	70.8	66.9	60.5	57.5
	2%	DB	52.5	50.6	55.3	63.4	73.6	80.7	84.7	83.5	78.7	71.0	63.1	57.3
		MCWB	50.8	47.3	47.6	52.0	62.2	69.4	73.3	73.3	70.0	65.5	59.3	54.7
	5%	DB	49.8	47.8	52.1	59.4	70.1	76.8	81.9	81.3	75.8	68.2	60.9	54.8
		MCWB	47.3	44.6	46.8	50.7	60.3	66.6	72.1	72.1	68.4	63.5	57.1	52.4
	10%	DB	46.2	44.9	49.1	56.6	66.2	73.6	79.3	79.2	73.5	65.8	58.3	51.9
		MCWB	43.1	41.1	44.9	49.2	58.1	65.8	70.8	71.0	67.3	61.9	54.9	49.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	51.4	55.2	59.7	67.7	74.3	77.9	78.2	75.0	69.8	63.0	58.5
		MCDB	54.7	52.2	59.0	66.3	76.0	82.5	86.0	83.8	79.5	72.0	64.0	60.0
	2%	WB	51.3	48.0	51.0	56.1	64.7	71.4	75.7	76.2	72.9	67.1	60.6	55.5
		MCDB	52.2	49.9	53.5	60.7	71.3	77.5	81.6	80.7	75.9	69.6	62.3	57.1
	5%	WB	47.6	45.2	48.1	53.4	62.3	69.7	74.3	74.7	71.2	64.9	58.1	53.1
		MCDB	49.2	47.0	50.9	56.2	67.5	74.4	79.3	78.8	73.7	67.1	60.4	54.6
	10%	WB	43.7	41.6	45.7	51.4	60.0	67.6	72.9	73.3	69.7	62.8	55.4	49.5
		MCDB	45.9	44.4	48.2	54.6	64.5	71.7	77.2	77.0	72.4	65.1	57.9	51.6
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.5	15.6	17.1	17.1	16.4	16.2	17.3	18.5	17.6	15.8	15.1
		MCDBR	16.1	18.0	18.4	22.5	22.0	20.3	18.0	17.5	18.6	18.2	15.5	16.2
		MCWBR	15.9	15.8	13.8	13.6	12.2	11.5	9.4	9.8	12.3	13.5	13.7	16.2
	5% WB	MCDBR	16.0	17.6	16.3	16.2	18.6	15.8	14.1	13.6	13.6	14.7	14.1	15.7
		MCWBR	16.3	17.0	14.6	12.3	11.4	10.4	8.2	8.2	10.0	12.7	14.2	16.5
Clear-Sky Solar Irradiance	taub	0.312	0.318	0.342	0.380	0.418	0.466	0.491	0.445	0.378	0.351	0.322	0.310	
	taud	2.502	2.490	2.459	2.377	2.312	2.215	2.153	2.277	2.440	2.520	2.559	2.552	
	Ebn at Noon	268	284	288	283	273	259	251	260	272	268	263	260	
	Edh at Noon	23	28	32	37	41	45	47	40	32	27	22	21	
All-Sky Solar Radiation	RadAvg	567	854	1182	1525	1712	1845	1898	1706	1390	950	640	482	
	RadStd	46	86	91	162	165	172	102	95	101	82	45	48	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	+86

Nomenclature: See separate page